

Fundamentals of Operating System Design

Module 5: Library Functions

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Learning Outcomes

- Define a library function.
- Explain the role of software libraries.
- Differentiate API, library function and system call.
- Explain how library functions interact with the kernel.

What is a Library Function?

Definition

A library function is a pre-written function provided by a software library to perform common tasks, allowing programmers to avoid rewriting code.

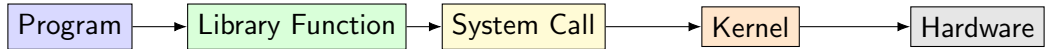
Examples:

- `printf()`
- `scanf()`
- `fopen()`
- `malloc()`
- `strlen()`

Why Use Library Functions?

- Reduce programming effort
- Reuse tested code
- Improve readability
- Increase portability
- Reduce development time

How a Library Function Works



API vs Library Function vs System Call

Feature	Library Function	System Call
Runs in	User Space	Kernel Space
Provider	C Library	OS Kernel
Purpose	Simplify programming	Request OS service
Example	printf()	write()

API is the interface used by applications. Library functions are API implementations that may internally invoke system calls.

Example: printf()

Execution Flow

Program → printf() → C Library → write() → Kernel → Display

- printf() formats the output.
- write() sends data to the operating system.
- The kernel writes data to the display device.

Examples of Common C Library Functions

Function	Purpose
printf()	Display output
scanf()	Read input
fopen()	Open file
fclose()	Close file
malloc()	Allocate memory
free()	Release memory
strlen()	Find string length
strcpy()	Copy string

Advantages of Library Functions

- Faster application development
- Reusable code
- Better reliability
- Platform independence
- Easier maintenance

Summary

- Library functions simplify programming.
- They execute in user space.
- Many library functions internally invoke system calls.
- System calls provide services from the kernel.

Quick Review

- ① What is a library function?
- ② Give four examples of library functions.
- ③ Differentiate API, library function and system call.
- ④ Explain how `printf()` reaches the kernel.